Definition and Properties of the Moment Generating Function

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1 Definition: Moment Generating Function

Let X be a random variable. The **Moment Generating Function (MGF)** of X is a function $M_X : \mathbb{R} \to [0, \infty]$ defined by:

$$M_X(t) = \mathbb{E}[e^{tX}]$$

for all $t \in \mathbb{R}$ for which the expectation is finite.

If X is a discrete random variable with probability mass function $p(x)$, this is:

$$M_X(t) = \sum_x e^{tx} p(x)$$

If X is a continuous random variable with probability density function $f(x)$, this is:

$$M_X(t) = \int_{-\infty}^{\infty} e^{tx} f(x)\, dx$$

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2 Why is it called "Moment Generating"?

The name comes from the fact that if you differentiate $M_X(t)$ at $t = 0$, you generate the **moments** of the random variable (its expected value, expected square, expected cube, etc.).

We can see this by expanding e^{tX} using its Maclaurin series:

$$e^{tX} = 1 + tX + \frac{(tX)^2}{2!} + \frac{(tX)^3}{3!} + \dots$$

Taking the expectation of both sides:

$$M_X(t) = \mathbb{E}[e^{tX}] = 1 + t\mathbb{E}[X] + \frac{t^2}{2!}\mathbb{E}[X^2] + \frac{t^3}{3!}\mathbb{E}[X^3] + \dots$$

Now, take the k-th derivative of $M_X(t)$ and evaluate it at $t = 0$:

- $M_X'(0) = \mathbb{E}[X]$ (the 1st moment, or mean)
- $M_X''(0) = \mathbb{E}[X^2]$ (the 2nd moment)
- $M_X^{(k)}(0) = \mathbb{E}[X^k]$ (the k-th moment)

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3 Key Properties

1. **Independence (The most important property for algorithms/ML):** If X and Y are independent random variables, then the MGF of their sum is the product of their MGFs:

$$M_{X+Y}(t) = \mathbb{E}[e^{t(X+Y)}] = \mathbb{E}[e^{tX}e^{tY}] = \mathbb{E}[e^{tX}]\mathbb{E}[e^{tY}] = M_X(t)M_Y(t)$$

   This property allows the multiplication of the MGFs of individual independent elements to obtain the MGF of the whole sum.

2. **Uniqueness:** If two random variables have the same MGF in an open interval around $t = 0$, they have the exact same probability distribution.

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4 Connection to the Chernoff Bound

The MGF is the fundamental tool of the **Chernoff Method**. The basic inequality states that for any $t > 0$,

$$\Pr(X \geq a) = \Pr(e^{tX} \geq e^{ta}) \leq \frac{\mathbb{E}[e^{tX}]}{e^{ta}} = \frac{M_X(t)}{e^{ta}}$$

By calculating the MGF, a problem about probabilities is transformed into a problem about algebra. One then chooses t to minimize the right-hand side, yielding an exponentially decaying bound on the tail probability.

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5 Examples of MGFs

- **Rademacher (coin flip, ± 1 with probability $1/2$):**

$$M_X(t) = \frac{1}{2}e^t + \frac{1}{2}e^{-t} = \cosh(t)$$

- **Normal Distribution $\mathcal{N}(\mu, \sigma^2)$:**

$$M_X(t) = \exp\left(\mu t + \frac{1}{2}\sigma^2 t^2\right)$$

- **Bernoulli(p):**

$$M_X(t) = (1-p) + pe^t$$

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6 Application to the Discrepancy Proof

To show that the discrepancy $D = \sum_{v \in S} \chi(v)$ is concentrated with high probability, one constructs its MGF. Because each $\chi(v)$ is a Rademacher variable, its MGF is $\cosh(t)$. Because the variables are independent, the MGF of the sum D is:

$$M_D(t) = \prod_{v \in S} \cosh(t) = (\cosh(t))^{|S|}$$

Using the inequality $\cosh(t) \leq e^{t^2/2}$, this becomes:

$$M_D(t) \leq \left(e^{t^2/2}\right)^{|S|} = e^{|S|t^2/2}$$

This is the MGF of a Gaussian. By plugging this into the Chernoff bound, one proves that the discrepancy is highly concentrated around 0 (with high probability), completing the proof.